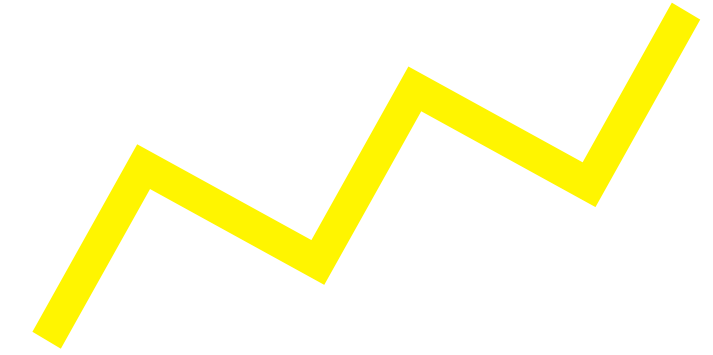
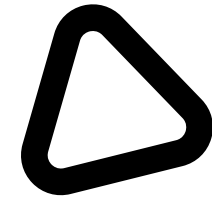




APPLIED MOTION-GRAPHICS

MTD.ba SS26

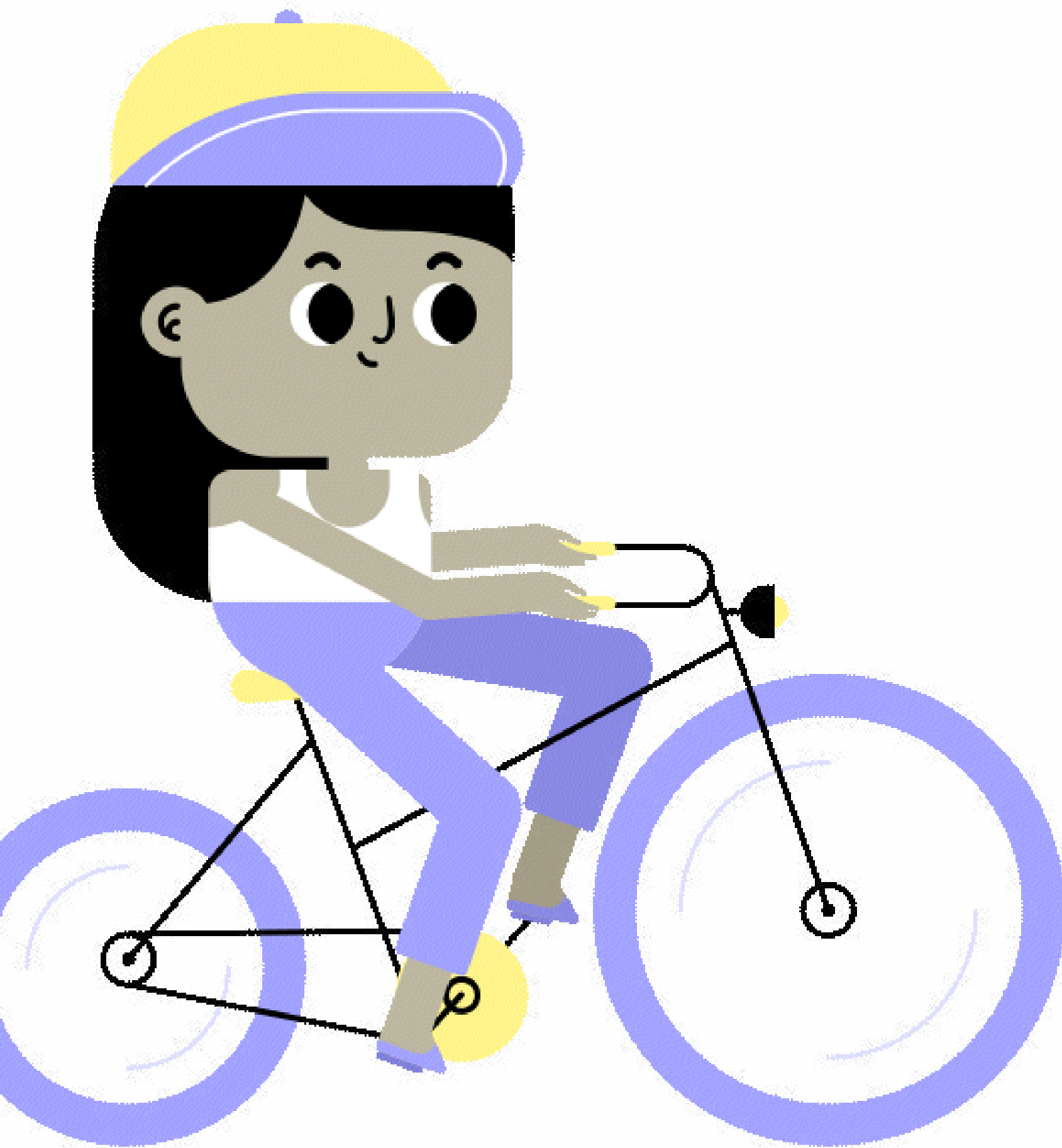
Character Animation

Adobe After Effects & Duik to JSON



4/5

Learning Objectives



Technical Rigging

→ Mastering **Bone structures** and **Inverse Kinematics** (IK) within After Effects for professional character control.

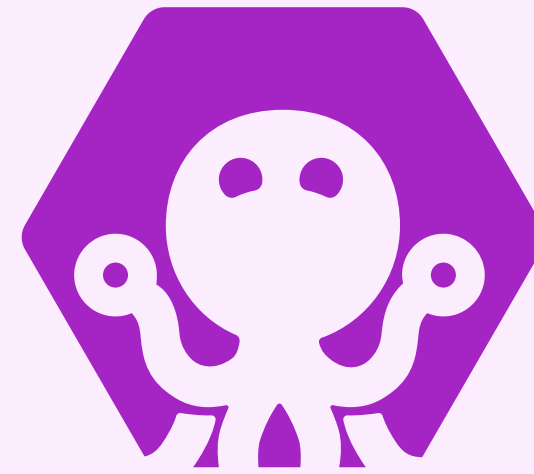
Lottie Logic

→ Understanding the Lottie/JSON Mindset: Designing character animation **optimized for web** and mobile performance.



Implementation

→ Learning the **pipeline** from vector design to interactive Framer prototypes using JSON export (Ae) and components (Framer).



AGENDA

- 1) **Basics**
- 2) **Preparation**
- 3) **Rigging**
- 4) **Animation**
- 5) **Export & Outlook**

Basics

Understanding Rigging



The Pipeline

1) Design

Vector data
(Illustrator)

→ Organizing
layers

2) Rigging

→ Duik Angela &
Auto-Rig Setup

+ animating path

3) Animate

Setting poses and
managing timing.

→ Curve editor

4) Export

Advanced Export
Settings

→ Expressions,
Guide Layers

5) Prototype

Insert different
states into Framer
composition.

→ short & sweet



Character Rigging Tools for After Effects

→ **Duik Ángela**

→ ~~Puppet Pin Tool (Ae)~~

→ Limber (versatile IK/FK)

→ Rubberhose (fast, cartoon-style)

→ Joysticks'n Sliders (facial rigging)

Duik Ángela

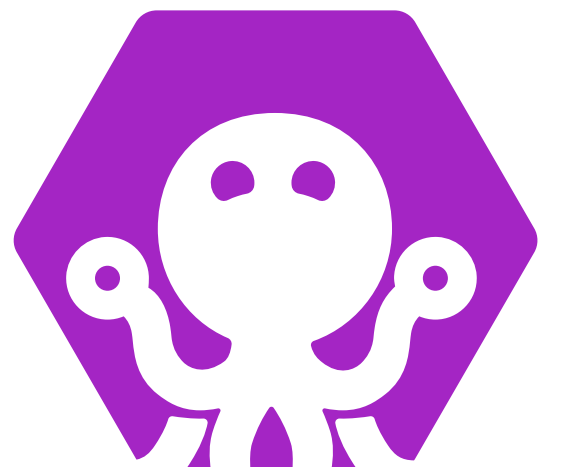
Duik Ángela is a comprehensive toolset for After Effects that brings professional animation features into a 2D environment.

→ Advanced **Rigging** (IK/FK/Bones)

→ Automated **Walk Cycles**

→ Efficient Keyframe Management

→ Free & **Open Source**



Rigging / Forward versus Inverse Kinematik



Forward Kinematics (FK)

A Rotation → B Rotation/Position → C Rotation/Position



Inverse Kinematics (IK)

C Position → B Position/Rotation → A Rotation

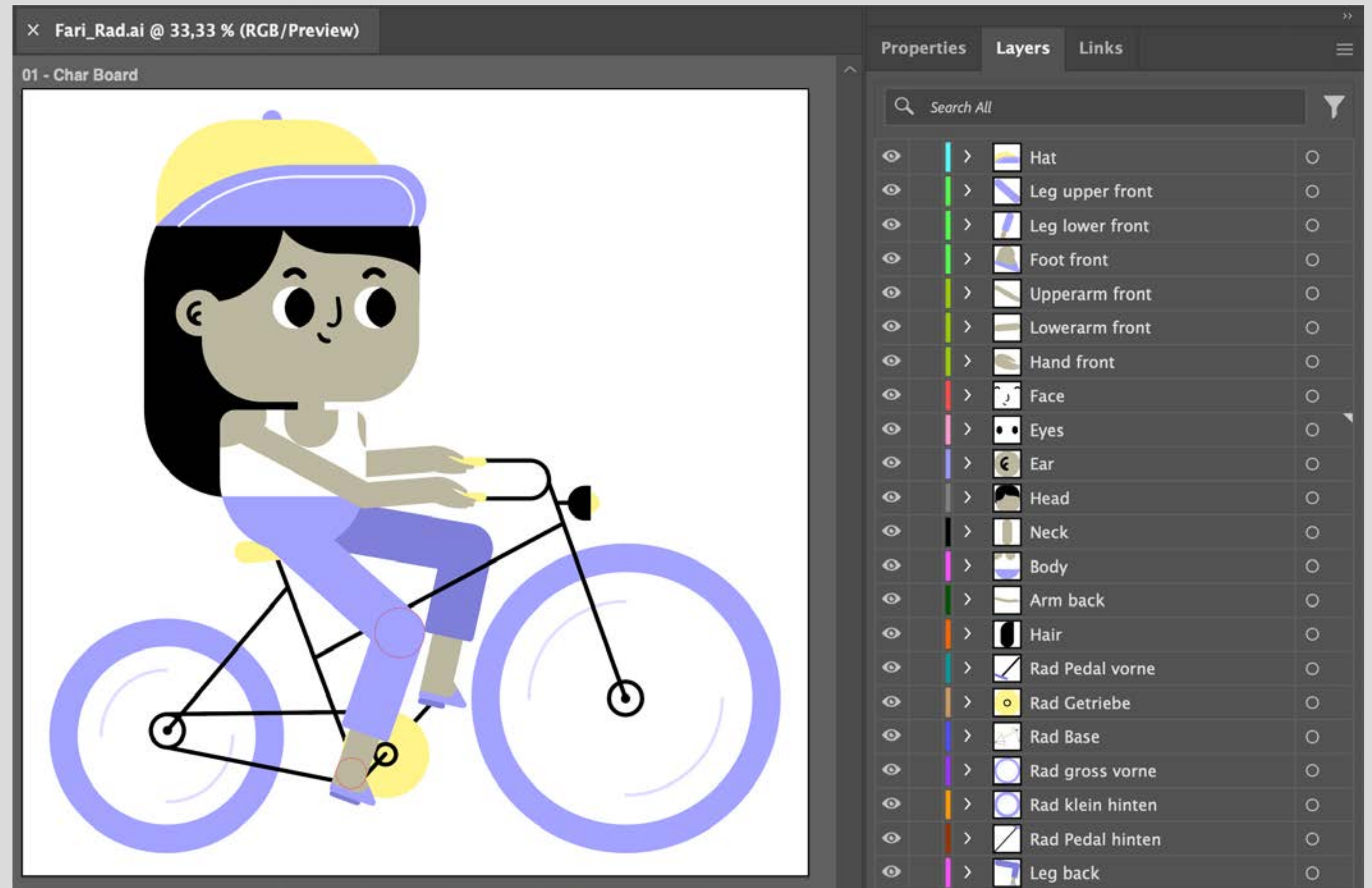
Preparation



Character Rigging Preparation is Key!

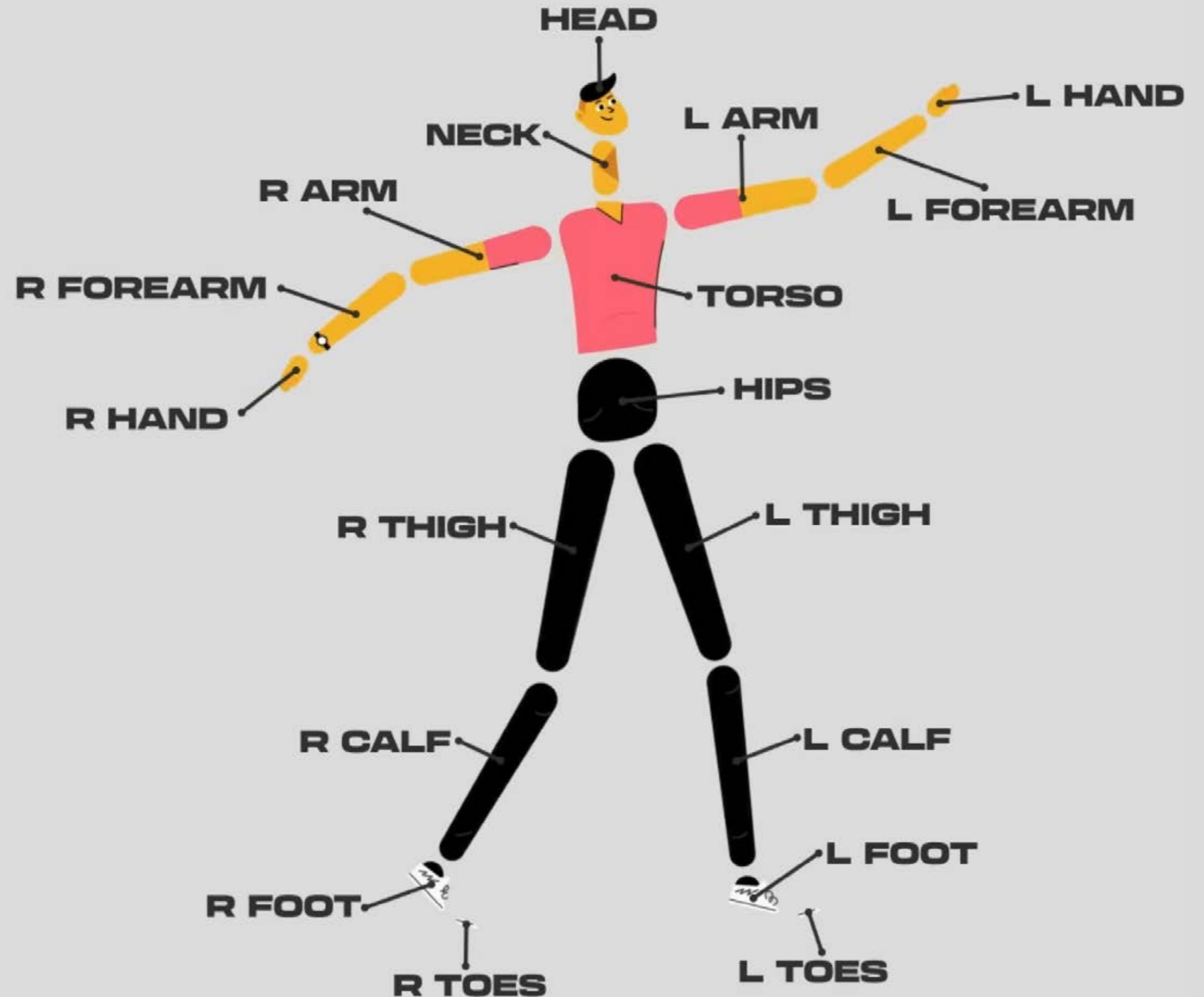
Adobe Illustrator

- Use **Separate Layers for every movable joint** (Hand, Forearm, Arm).
- **Avoid:** merged shapes, grouped shapes that you want to animate individually.



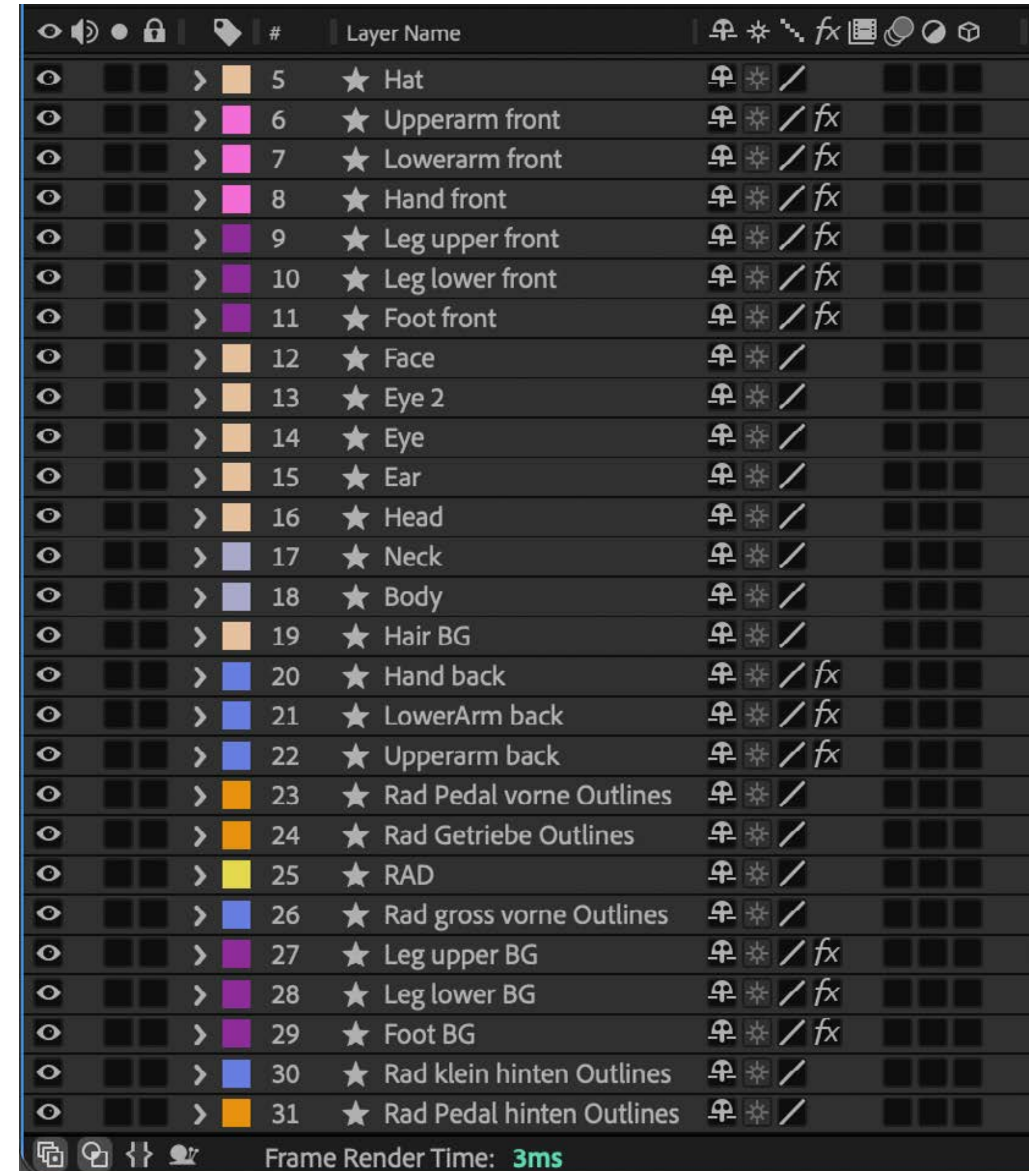
Character Rigging Preparation is Key!

- **Naming:** always name/label your layers, there will be many!
- **Overlap:** Ensure limb joints overlap properly to avoid gaps during rotation.
- **Clean shapes:** no filters, no complex blends, no extra points.



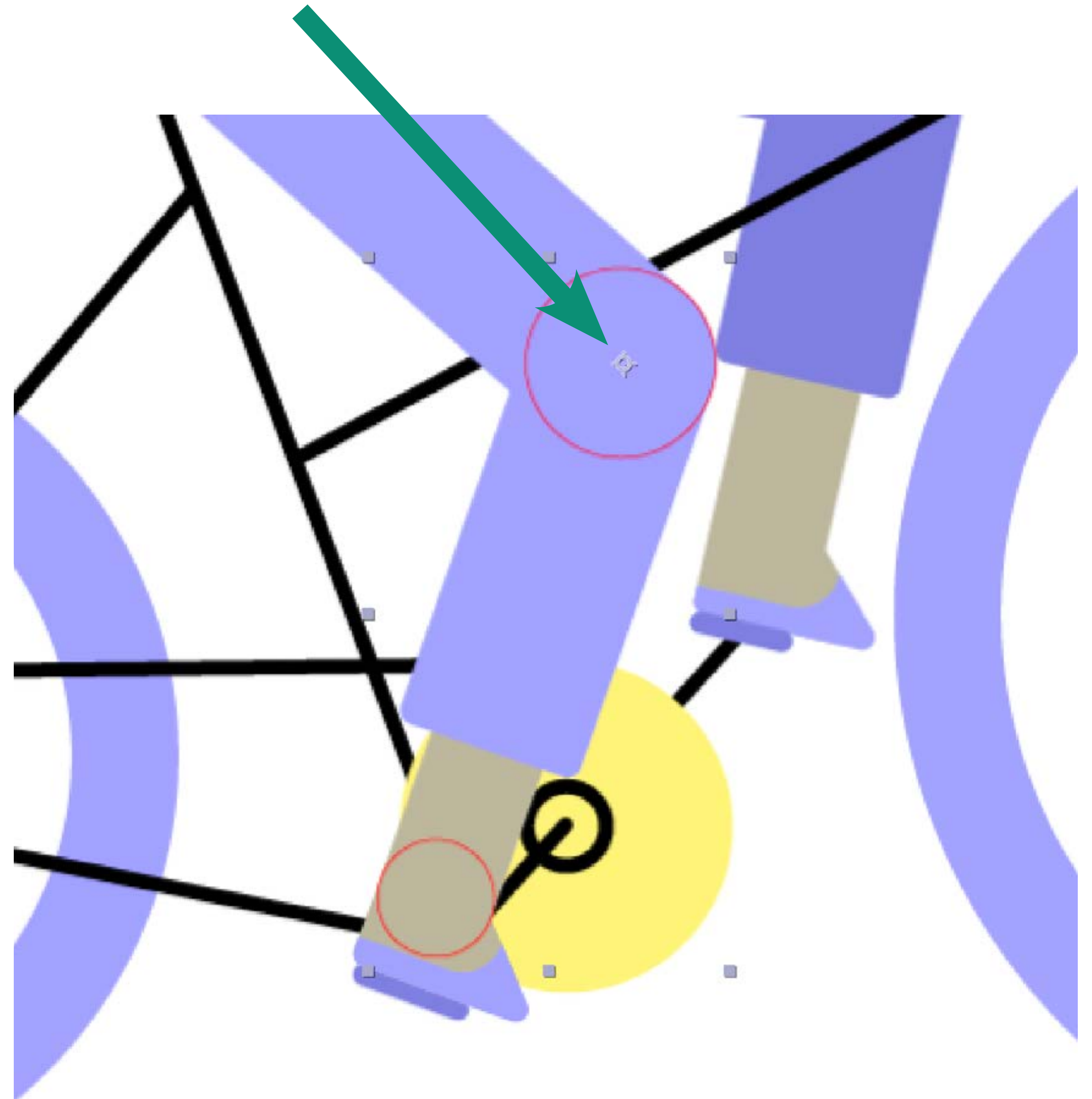
After Effects: Work with Shapes

- Import as: **Composition – Retain Layer Sizes**
- Lottie/JSON cannot read Illustrator (.ai) files. Every layer must be converted into a native After Effects **Shape Layer**.
- Shortcut: Right Click > Create > Shape from Vector Layer
- Delete all .ai layers from your composition (Bodymovin Export).



Pivot Points: Set the anchor right!

- The **Anchor Point** is the most critical element in rigging.
- Place anchor points **exactly** at the **center of the joint rotation** before starting any Duik operations!
- Create **guides** in your design.
- **Test the rotation** of every limb.



Parenting in After Effects

Parent everything in right order.

- Foot to Lower Leg to Upper Leg to Hip to Body
- Face (& every Facial Detail) to Head to Neck to Body
- Toe or Tip of Shoe to Shoe / Heel to Lower leg to Upper Leg to Hip to Body

Why this matters later:

- Bones, IK, and automated rigs rely on predictable hierarchies.

#	Layer Name	Parent & Link
5	Hat	16. Head
6	Upperarm front	18. Body
7	Lowerarm front	6. Upperarm
8	Hand front	7. Lowerarm
9	Leg upper front	18. Body
10	Leg lower front	9. Leg upper
11	Foot front	10. Leg lower
12	Face	16. Head
13	Eye 2	16. Head
14	Eye	16. Head
15	Ear	16. Head
16	Head	17. Neck
17	Neck	18. Body
18	Body	25. RAD
19	Hair BG	16. Head
20	Hand back	21. LowerAr
21	LowerArm back	22. Upperar
22	Upperarm back	18. Body
23	Rad Pedal vorne Outlines	24. Rad Geti
24	Rad Getriebe Outlines	25. RAD
25	RAD	None
26	Rad gross vorne Outlines	25. RAD
27	Leg upper BG	18. Body
28	Leg lower BG	27. Leg uppi
29	Foot BG	28. Leg lower
30	Rad klein hinten Outlines	25. RAD
31	Rad Pedal hinten Outlines	24. Rad Geti

Rigging



Understanding the Bone Structure

Bones are non-rendering guide layers that act as the **structural framework** of your character.

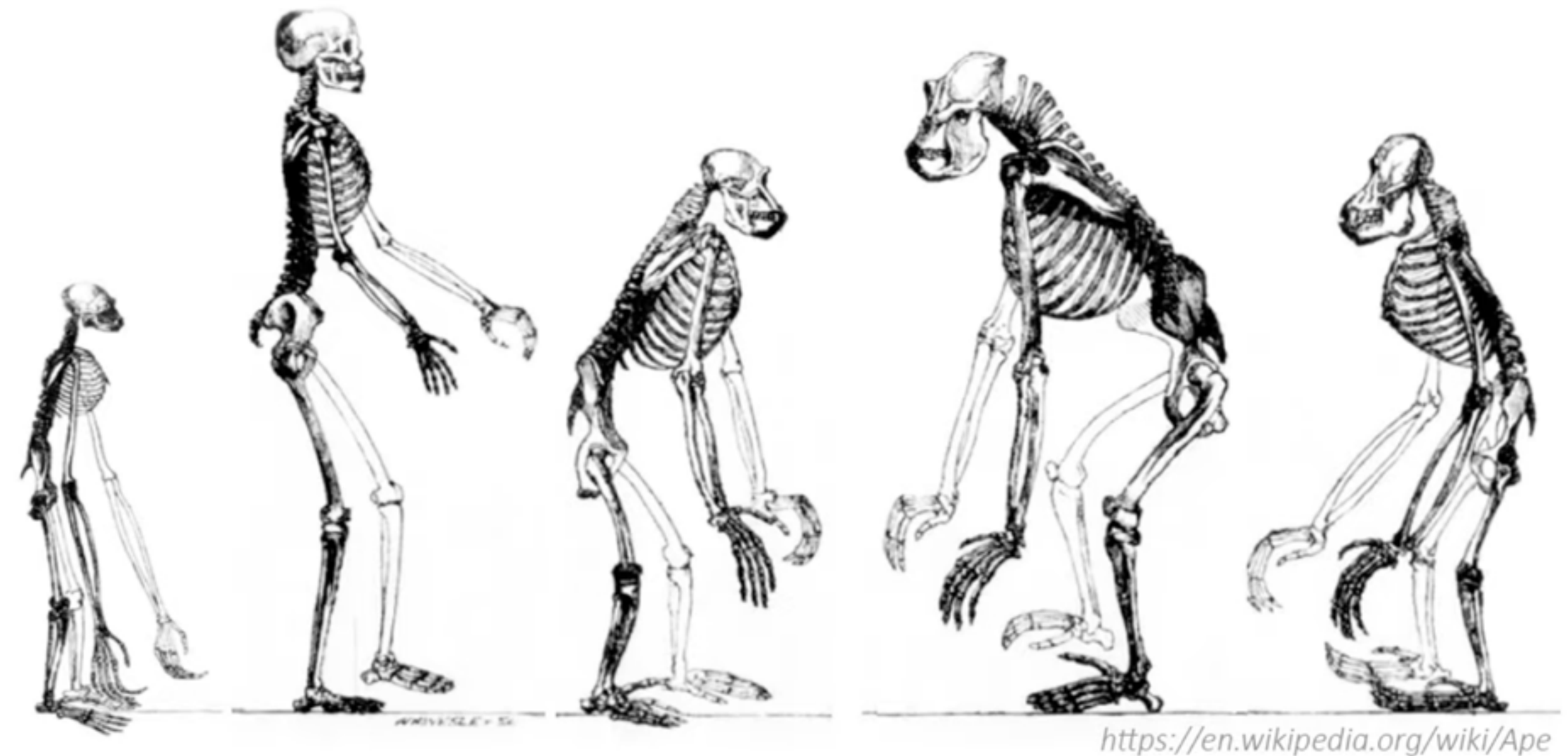
They handle the complex mathematical calculation (IK), allowing your visual layers to follow along simply via Parenting.

Why IK for characters?

IK gives you professional control over: arms, legs, spines, tails.

→ Duik Angela automates this inside Ae.

→ bones = joints,

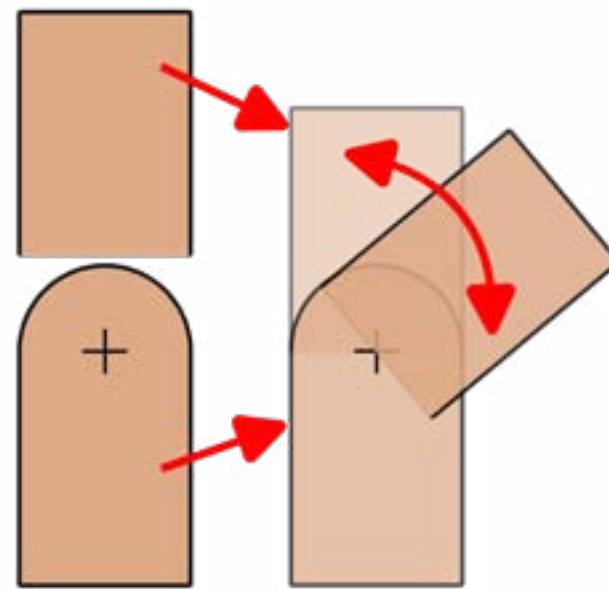


→ IK controllers = the “handles” you move,

→ FK (Forward Kinematics) = moving each joint by hand,

→ IK (Inverse Kinematics) = move the hand → arm/forearm follow automatically.

Mechanical Joint Placement



Anatomy & Parenting

Place Bones at physiological centers: Shoulder, Elbow, Wrist. Test with Rotation!

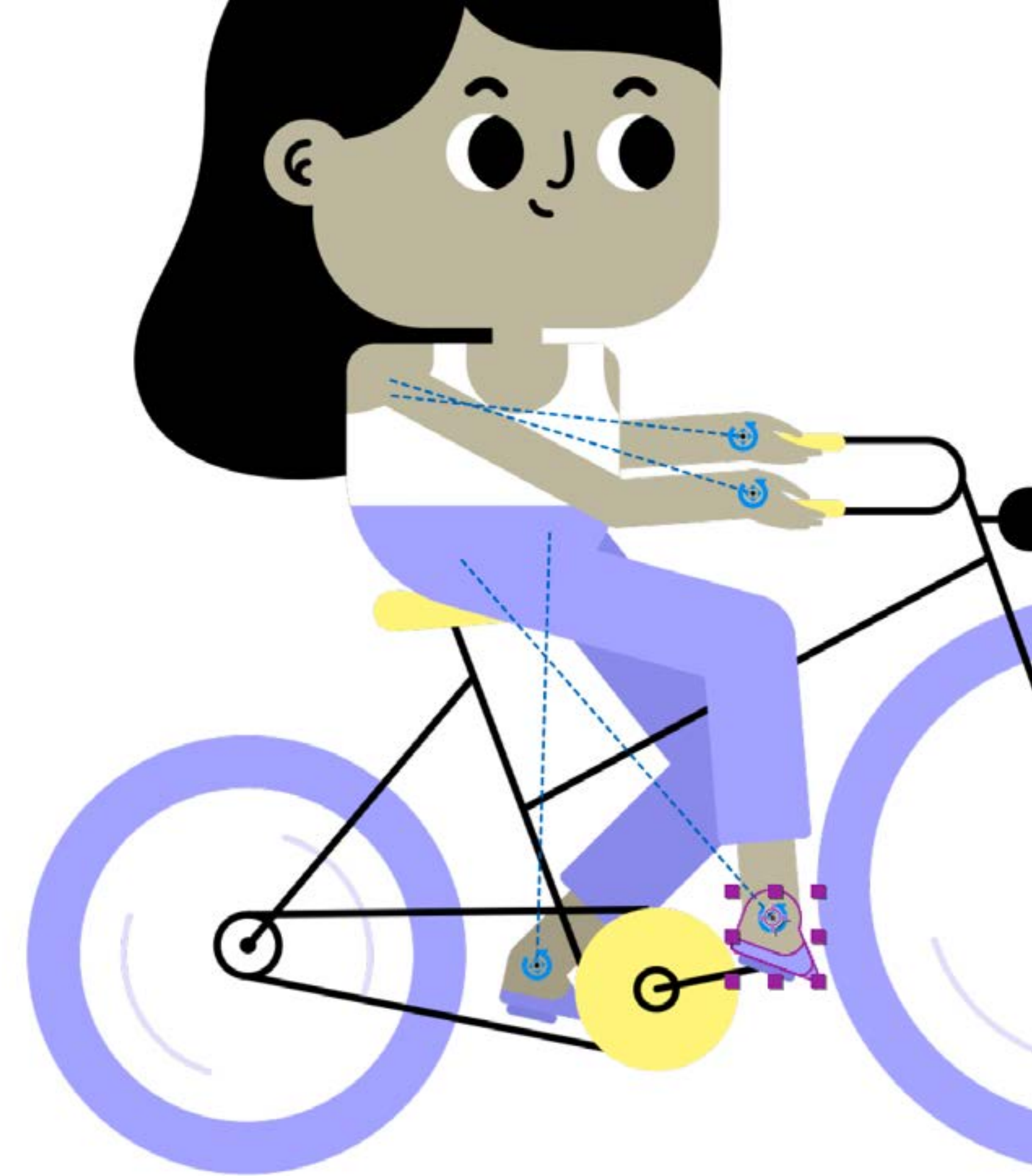
Parent each shape element to its parent (the left foot to lower left leg and so on).

Generating the Rig

Bones must be parented to the visual layer they represent.

Select all chained elements.

→ **generate Auto Rig**

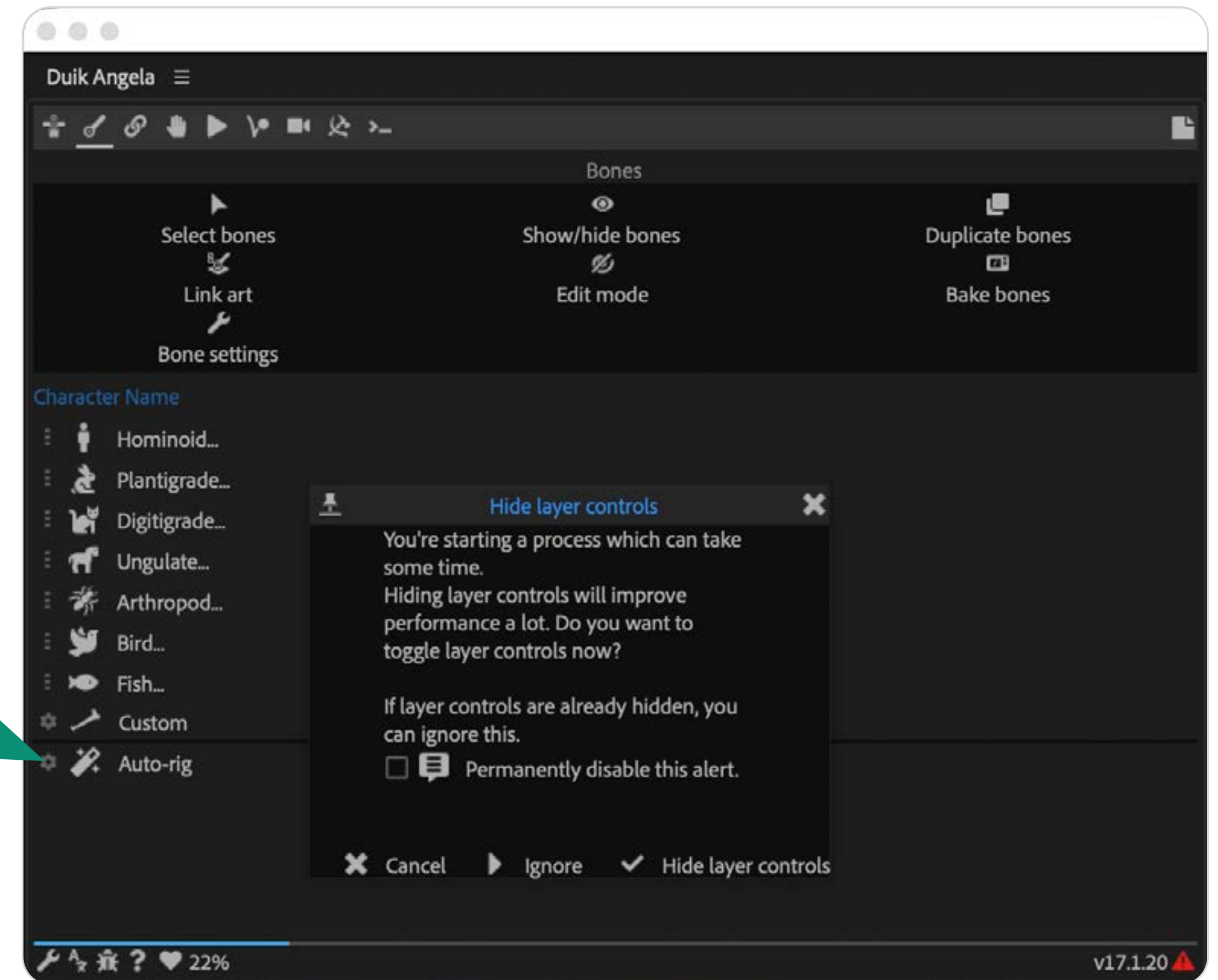


The Hinge

Now each Graphic Layer is parented to the Duik Structure (lock the graphic for animation).

Auto Rig: The magical Button

1. In the comp: select your vector layers (upper arm, lower arm, hand).
2. In Duik menu: create an Arm structure or Humanoid structure (if you have different / extended limb-construction, else: use Auto rig).
 - Then: Snap the pieces (arm/forearm/hand) into the matching rig parts.
3. Use **Auto-Rig** to parent artwork to structure bones.
4. **Start animating:** move the controllers!
 - Everything still works inside After Effects, but Duik now handles IK / FK logic.



IK vs. FK



Inverse Kinematic

Use IK when you want to control the **end effector** (hand, foot, tail tip, neck end) and let the chain “solve itself” automatically.

1. Character stands on the ground, walks, or steps
2. Hands lie on surfaces or objects
3. Climbing, hanging, gripping
4. Sitting, leaning, crouching poses
5. Precise placement of hands/feet

Movement starts with contact, then lifts: IK → FK

Forward Kinematic

Use FK when you want organic, **flowing, arc-like motion** and control the movement from the “root” outward (shoulder → elbow → hand, etc.).

1. Flowing, natural arm movements
2. Sword swings, punches, throws
3. Expressive, gestural body language
4. Spine, neck, tail, hair, antennas
5. Movement with no world contact

Movement starts freely, then contacts: FK → IK

Facial Rig Basics

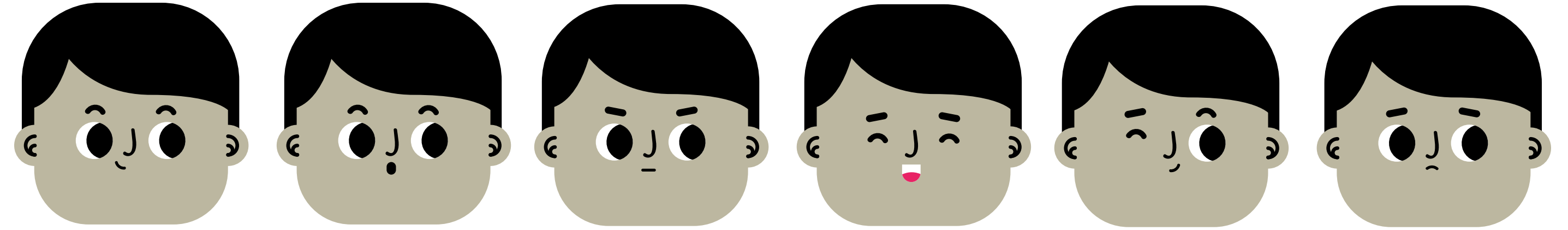
Many characters live only in the face.

Rigging idea: No complex Duik Puppet-Rig, no bones.

Only separate mouth, eyes, brows layers with position/opacity/shape changes.

Techniques:

- Morphing shapes for different mouth states,
- Eyelid layers for “open/closed/blinking”,
- Brow layers for “neutral, raised, skeptical”.



Prepare for States

Define 2-4 key states in Ae:

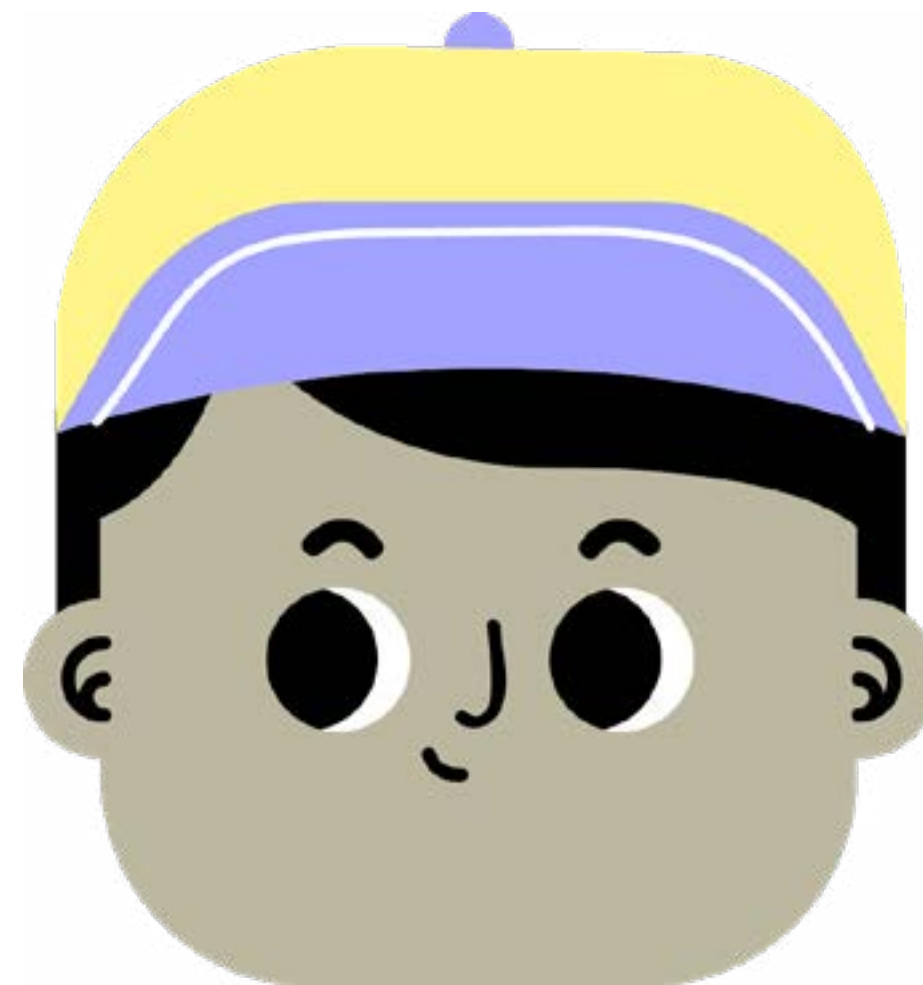
- neutral, smile, surprise, blink.
- Each state lives on its own layer or as different keyframes in a sub-comp.

Export later as:

- a single looping Lottie for a neutral smile,
- or separate short animations for different Framer States.

Lottie-Logic hint:

- keep face states small, short, and without complex effects.



Animation



Keyframe animation

Minimize

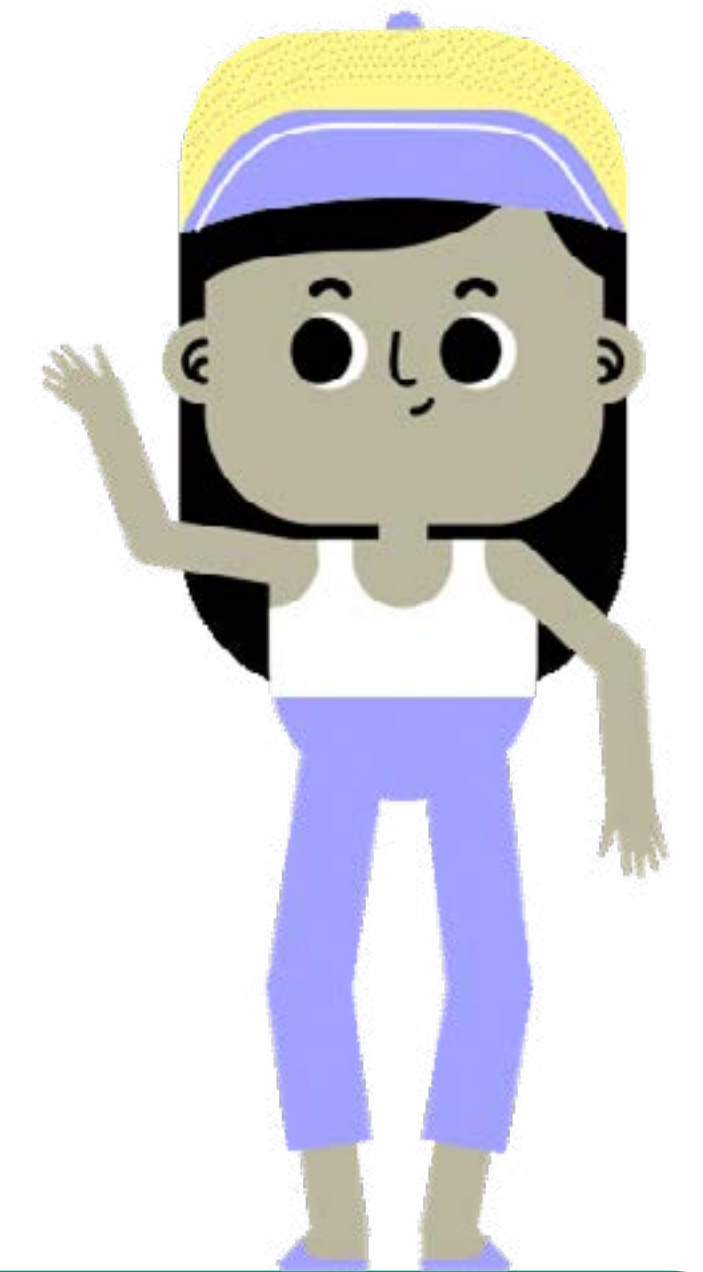
Every keyframe adds data to the JSON output. Use as few as possible!

Ease

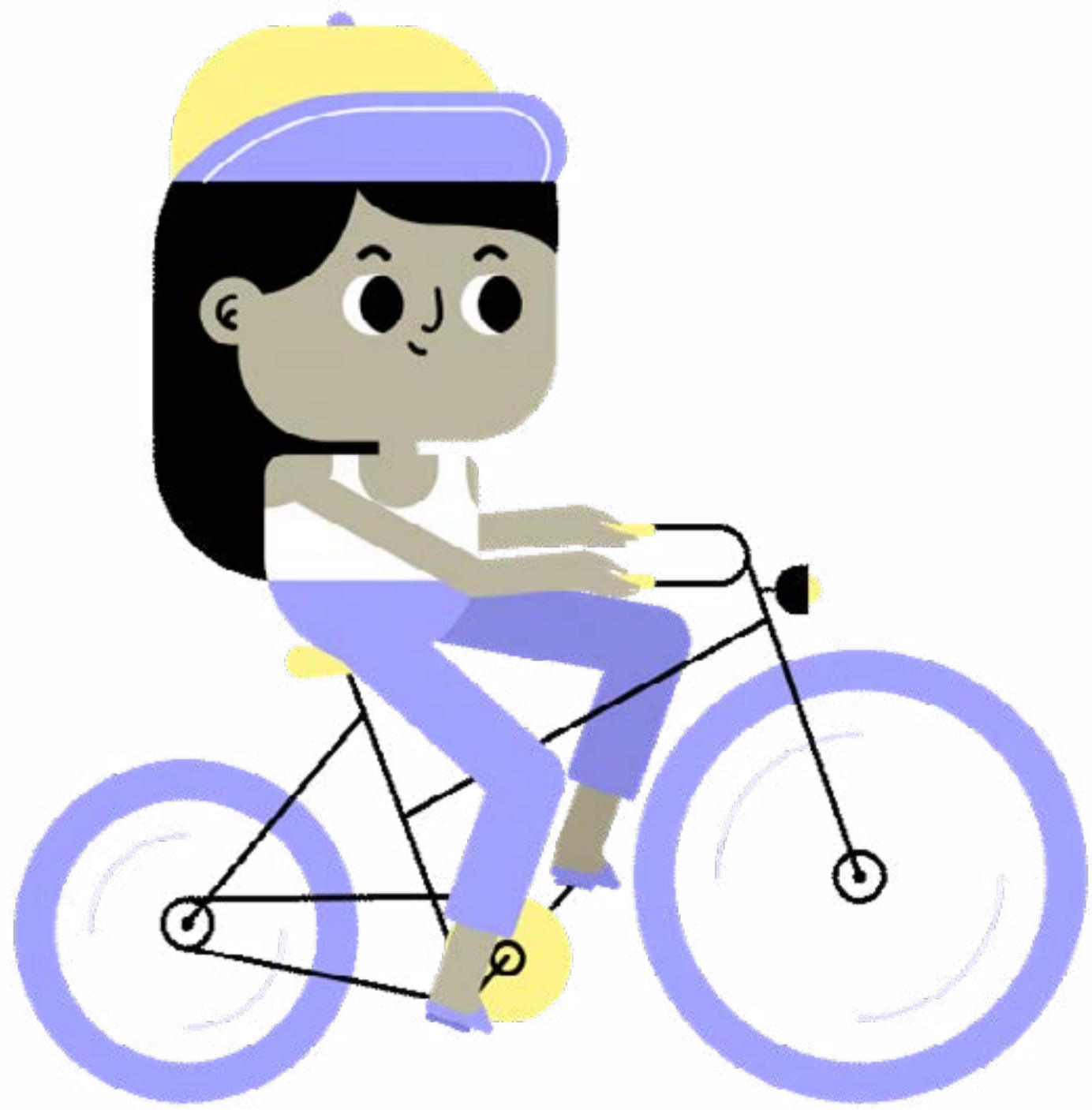
Activate Bezier curves and edit them in the Curve editor to generate very unique transitions.

Timing

Keep Game like animations under 2 seconds for snappiness!



Generate short animations for your Framer prototype



Export & Outlook



Bodymovin

Advanced Settings

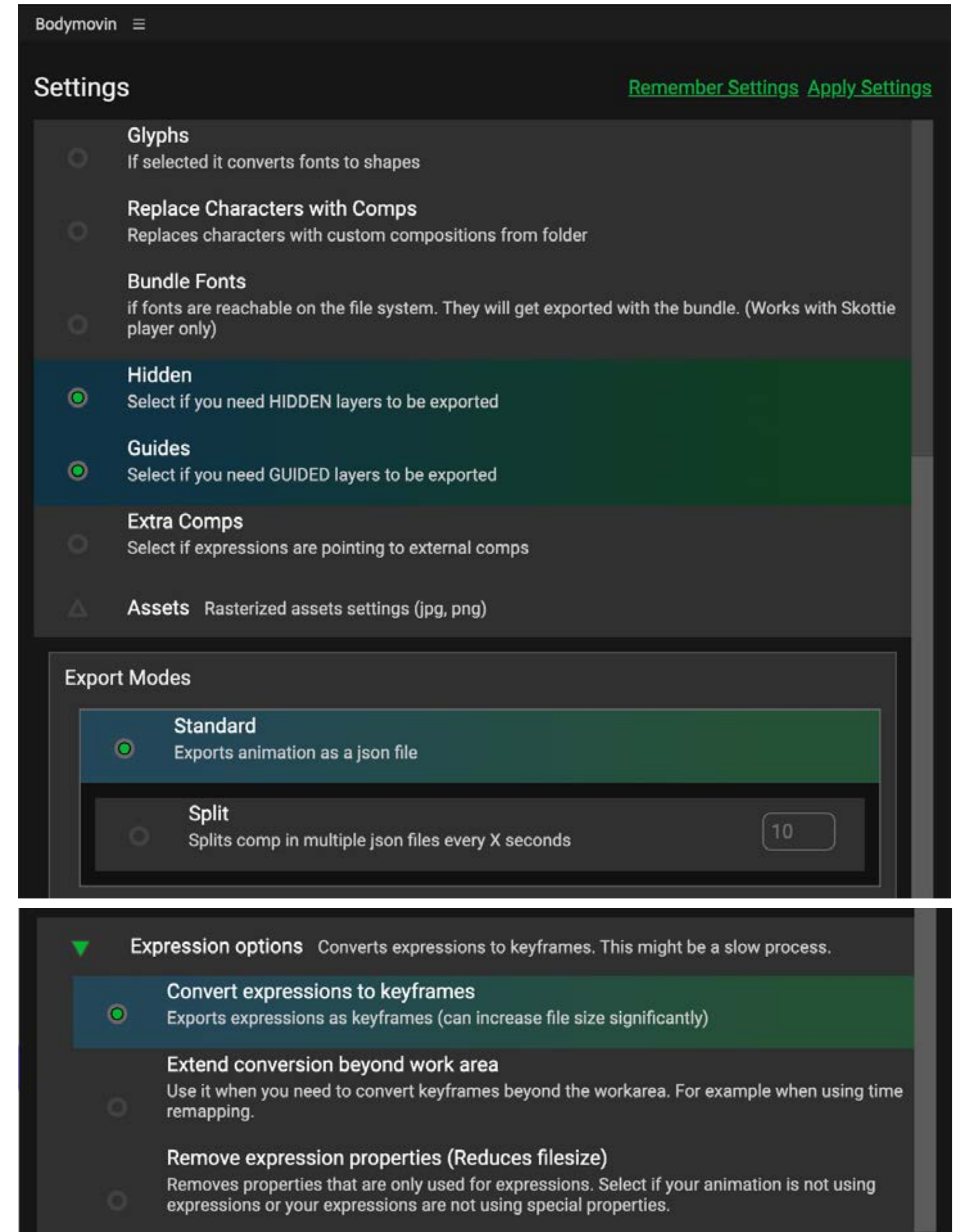
With complex rigging you have to make some more Settings in Bodymovin before exporting:

→ **Hidden** (you have hidden layers)

→ **Guides** (you have layers which guide your rig)

Expression options

→ **Convert expressions to keyframes** (bigger file size!)



Exercise Block 4:

Outlook

First steps in Rigging

Rigging course:

- Facial Rig
- Duik Body Rig
- IK versus FK

Animate different figures

Animate the rigged characters and export JSON.

Key elements:

- Character Animation

Animate your own Character

Animate your own character for your prototype or continue your work in **Framer**.

